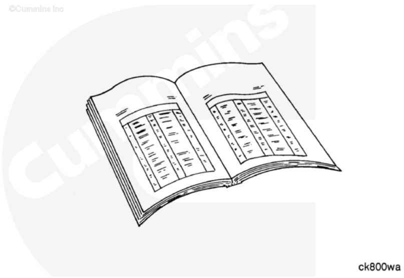
Trainings

Clean and Inspect for Reuse

Clean and inspect the connecting rod for reuse. Refer to Procedure 001-014 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-014-tr.html)

Clean and inspect the pistons for reuse. Refer to Procedure 001-043 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-043.html)



Install

⚠ CAUTION ⚠

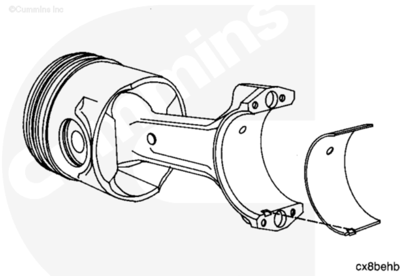
To reduce the possibility of engine damage, do not lubricate the back side of the bearing shells.

Clean the connecting rod and the bearing shells with a lint-free cloth.

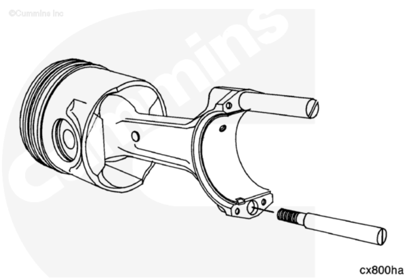
Install the rod bearing. Be sure the tang is positioned as shown. The end of the bearing **must** be even with the cap mounting surface.

Use clean engine oil. Lubricate the bearing surface. The bearings **must** be installed in their original locations if new bearings are **not** used.

Note : All of the rod bearings are identical.

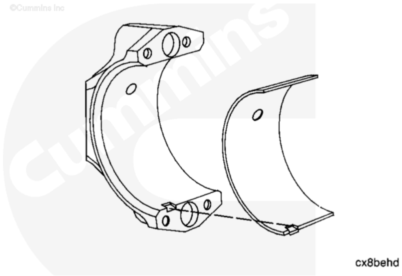


Install two connecting rod guide pins, Part Number 3375098, in the connecting rod. The guide pins will aid the assembly process and protect the crankshaft.



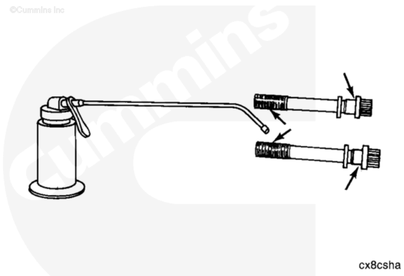
⚠ CAUTION ⚠

The connecting rods and connecting rod caps are not interchangeable. The connecting rods and the connecting rod caps are machined as an assembly. Engine damage will result if they are mixed.

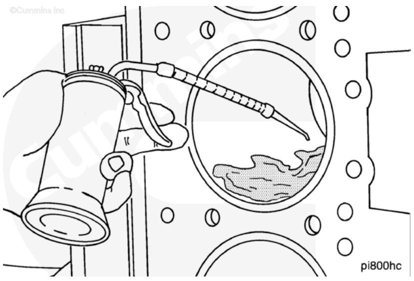


Install the lower bearing shell in the rod cap. Be sure the tang of the bearing shell is in the slot of the cap and the end of the bearing is even with the cap surface.

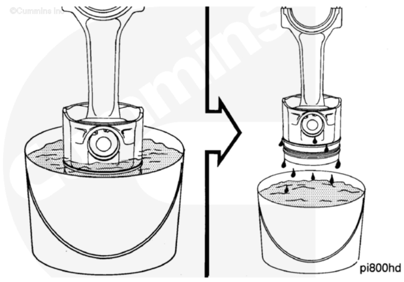
Use clean engine oil. Lubricate the connecting rod capscrews and washers as shown. Install the washers and capscrews in the caps.



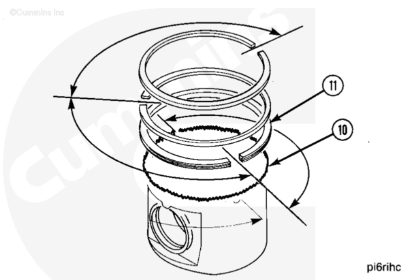
Lubricate the cylinder liner with clean engine oil. The entire bore **must** be lubricated.



Immerse the piston in clean engine oil until the rings are covered. Allow the excess oil to drip off the assembly.

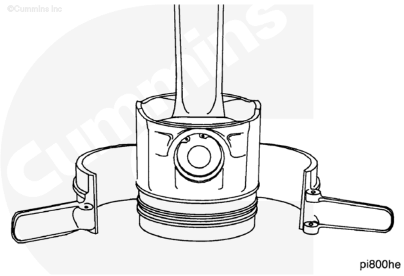


Check to be sure the ring gap position is still correct.



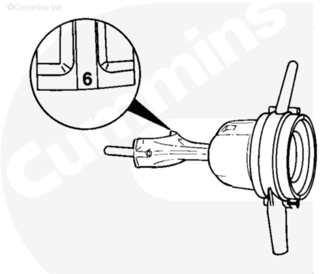
⚠ CAUTION ⚠

To reduce the possibility of engine damage, make sure the piston rings fit correctly in the piston.


pi800he

Use a piston ring compressor, Part Number 4918898, or equivalent. Install the ring compressor on the piston.

Note : The ring compressor has a tapered bore. The small end of the taper must be positioned toward the piston skirt.



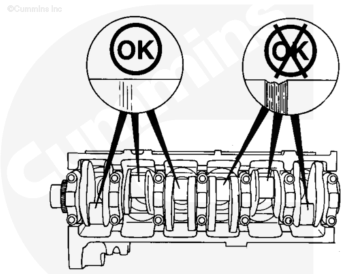
pi800hf

The cylinder number on the connecting rod and connecting rod cap **must** be the same.

The side of the rod with the cylinder number (bearing tang side) **must** be toward the camshaft.

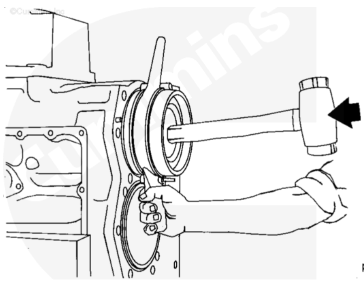
Rotate the crankshaft until the journal for the rod being installed is at BDC.

Note : If the engine has a crankshaft with bolt-on counterweights, the journal must be at top dead center (TDC).



ks400sa

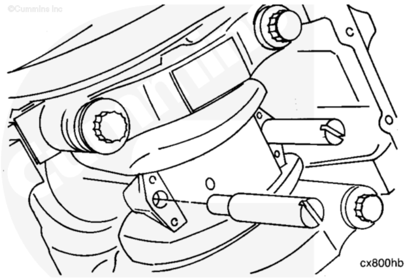
Check the crankshaft rod journals for damage.


pi400ha

Install the rod and piston until the ring compressor touches the block. Align the rod with the crankshaft journal.

Hold the ring compressor firmly against the block. Use a wooden hammer handle to push the piston into the liner.

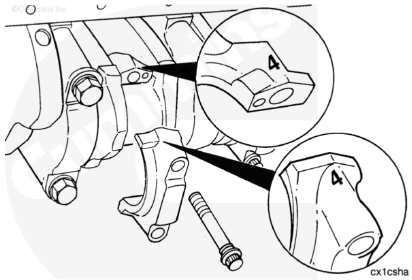
Push the piston into the bore until the rod bearing contacts the crankshaft journal. Remove the guide pins.



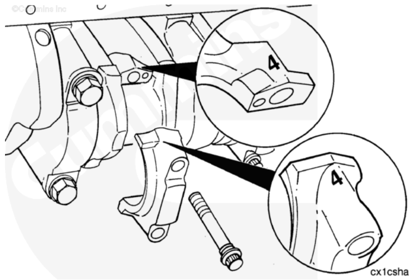
The cylinder number on the rod and cap **must** be the same.

The side of the cap with the cylinder number marking (bearing tang side) **must** be toward the camshaft.

Install the connecting rod cap.



Tighten the capscrews alternately and evenly to pull the cap over the dowel pins. Use the following steps to tighten the capscrews.

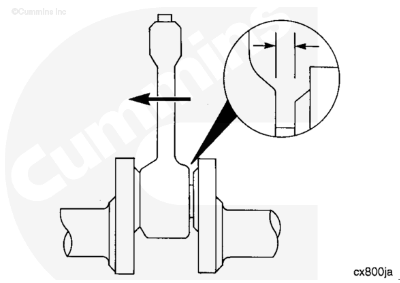


Torque Value:

1. 102 n•m [75 ft-lb]
2. 197 n•m [145 ft-lb]
3. 292 n•m [215 ft-lb]
4. 366 n•m [270 ft-lb]
5. Loosen both capscrews to remove all tension.
6. 102 n•m [75 ft-lb]
7. 149 n•m [110 ft-lb]

Plus 90° to achieve correct bolt stretch.

Check the side clearance between the rod and the crankshaft. The rod **must** move freely from side-to-side.



Measurements over 0.51 mm [0.020 in] **must** be measured with a dial indicator.

If the side clearance is greater than 0.51 mm [0.020 in] do **not** reuse the parts unless the same connecting rod is used on the same crankshaft journal in the same engine. If the parts are reused on the same crankshaft connecting rod journal in the same engine, the maximum side clearance measured with a dial indicator is 5.08 mm [0.200 in]

Connecting Rod and Crankshaft Side Clearance, New or Remanufactured Parts		
mm		in
0.20	MIN	0.008
0.35	MAX	0.014

Connecting Rod and Crankshaft Clearance, Used Parts		
mm		in
0.51	MAX	0.020

Connecting Rod and Crankshaft Clearance, Same Connecting Rod and Crankshaft Reused on Same Journal of Crankshaft		
mm		in
5.08	MAX	0.200

Last Modified: 11-Aug-2016